

$$= \frac{e^{iKx} - e^{-iKx}}{iKx} = \frac{2 \sin(Kx)}{Kx}$$

Thus,

$$\tilde{V}(K) = 2\pi \times (-\alpha) \int_{-\infty}^{\infty} dx \left(\frac{x \sin(Kx)}{Kx} \right)$$

$$= -\frac{4\pi\alpha}{K} \int_0^{\infty} \sin(Kx) dx$$

This integral is diverging, so we regularize it using Yukawa form $e^{-\mu x}/x$ with $\mu \rightarrow 0^+$:

$$\int d^3x e^{-\vec{K} \cdot \vec{x}} \frac{e^{-\mu x}}{x} = \frac{4\pi}{K^2 + \mu^2}$$

Here the needed part is:

$$\begin{aligned} 4\pi \int_0^{\infty} dx \, x \frac{\sin(Kx)}{Kx} e^{-\mu x} &= \frac{4\pi}{K} \int_0^{\infty} dx \sin(Kx) e^{-\mu x} \\ &= \frac{4\pi}{K} \cdot \frac{K}{\mu^2 + K^2} = \frac{4\pi}{\mu^2 + K^2} \end{aligned}$$

Now, by using $\int_0^{\infty} \sin(Kx) e^{-\mu x} dx = \frac{K}{\mu^2 + K^2}$ & taking the limit $\mu \rightarrow 0$ gives;

$$\tilde{V}(K) = -\frac{4\pi\alpha}{K^2 + \mu^2} \xrightarrow{\mu \rightarrow 0} \boxed{\tilde{V}(K) = -\frac{4\pi\alpha}{K^2}}$$

(2.)

We have;

$$k^2 \rightarrow k^2 + m^2$$

$$\tilde{V}(k) = -\frac{4\pi\alpha}{k^2 + m^2}, \text{ \& } 4\pi \text{ come from the}$$

Standard 3-D Fourier transform convention in natural units ($\hbar = c = 1$) for the Coulomb potential, $\tilde{V}(k) = -\frac{4\pi\alpha}{k^2}$

We'll perform inverse Fourier transform, to find the co-ordinate-space potential, $V(r)$.

$$V(r) = \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k} \cdot \vec{r}} (\tilde{V}(k))$$

Due to spherical symmetry;

$$V(x) = \frac{-\alpha}{2\pi^2 x} \int_0^{\infty} dk \frac{(k \sin(kx))}{k^2 + m^2} \cdot 4\pi$$

$$I = \frac{-2\alpha}{\pi x} \int_0^{\infty} dk \frac{(k \sin(kx))}{k^2 + m^2}$$

we'll find this integral using Contour Integration:

$$I = \int_0^{\infty} \frac{k \sin(kx)}{k^2 + m^2} dk, \quad x > 0 \text{ and } m > 0$$

Let;

$$J = \int_{-\infty}^{\infty} \frac{k e^{ikx}}{k^2 + m^2} dk$$

simple poles at $k = \pm im$

For $x > 0$, the e^{ikx} decays for $\text{Im}(k) > 0$ so we close the contour there, enclosing the pole at $k = im$.

The residue at $k = im$ is

$$\text{Res}_{k=im} = \left(\frac{k e^{ikx}}{(k-im)(k+im)} \right) = \frac{(im) e^{i(im)x}}{(im+im)}$$

$$= \frac{im e^{-mx}}{2im} = \frac{e^{-mx}}{2}$$

$$\therefore J = (2\pi i) \cdot \left(\frac{e^{-mx}}{2} \right) = \pi i e^{-mx}$$

Now, the integrand $\frac{k e^{ikx}}{k^2 + m^2}$ has real part $\frac{k \cos(kx)}{k^2 + m^2}$ (odd)

and imaginary part $\frac{(k \sin(kx))}{k^2 + m^2}$ (even funcⁿ)

So, the integral over the odd part in $(-\infty, \infty)$ vanishes.

$$\text{Thus; } J = i \int_{-\infty}^{\infty} \frac{k \sin(kr)}{k^2 + m^2} dk = i \cdot 2I$$

$$\therefore i \cdot 2I = \pi i e^{-mr}$$

$$\Rightarrow \boxed{I = \frac{\pi}{2} e^{-mr}}$$

So,

$$V(r) = \frac{-2\alpha}{\pi r} I = \frac{-2\alpha}{\pi r} \times \frac{\pi}{2} e^{-mr}$$

$$\boxed{V(r) = -\frac{\alpha}{r} e^{-mr}} \rightarrow \text{Yukawa potential}$$

Also,

$$m = 140 \text{ MeV}, \quad r = 1 \text{ fm} \quad \text{and} \quad \alpha_s = 0.2$$

$$\therefore \hbar c \approx 197.3 \text{ MeV} \cdot \text{fm}$$

$$mr \approx \frac{mr \hbar c}{\hbar c} = \frac{140 \text{ MeV} \cdot \text{fm}}{197.3 \text{ fm} \cdot \text{MeV}} \approx 0.709$$

$$\text{So; } e^{-0.709} \approx 0.492$$

$$\left(\because r = 1 \text{ fm} = \frac{1}{197.3 \text{ MeV}} \right)$$

$$\frac{\alpha_s}{r} = (0.2)(197.3 \text{ MeV})$$

$$= 39.46 \text{ MeV}$$

$$\text{So, } V(r) = -(39.46 \text{ MeV}) \times 0.492$$

$$\approx -19.4 \text{ MeV} \quad \underline{\underline{\text{Ans}}}$$

④

Given dual tensor :

$$G_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\nu\alpha\beta} F^{\alpha\beta}, \rightarrow (i)$$

& Maxwell's eqⁿ. in Vacuum is;Levi-Civita symbol
with $\epsilon_{0123} = +1$

$$\partial^\nu G_{\mu\nu} = 0 \rightarrow (ii)$$

put $G_{\mu\nu}$ in (ii):

$$\partial^\nu \left(\frac{1}{2} \epsilon_{\mu\nu\alpha\beta} F^{\alpha\beta} \right) = 0$$

$$\Rightarrow \frac{1}{2} \epsilon_{\mu\nu\alpha\beta} \partial^\nu F^{\alpha\beta} = 0$$

$$\Rightarrow \epsilon_{\mu\nu\alpha\beta} \partial^\nu F^{\alpha\beta} = 0$$

For clarity we relabel the dummy indices, $\alpha \rightarrow \rho$ & $\beta \rightarrow \sigma$;

$$\therefore \boxed{\epsilon_{\mu\nu\rho\sigma} \partial^\nu F^{\rho\sigma} = 0} \quad (\forall \mu, \rho \& \sigma) \rightarrow (iii)$$

 $\epsilon_{\mu\nu\rho\sigma}$ is totally anti-symmetric in all indices.

$$\Rightarrow \partial^\nu F^{\rho\sigma} \text{ is anti-symmetric in } (\rho\sigma) \text{ because } \boxed{F^{\rho\sigma} = -F^{\sigma\rho}}$$

So, the contraction in (iii) vanishes for all μ , i.e. totally anti-sym part, in the indices $\rho\sigma$, of $\partial^\nu F^{\rho\sigma}$ is zero

For a tensor $A^{\alpha\beta\gamma}$ anti sym in $\beta\gamma$,
the condition $\epsilon^{\alpha\beta\gamma\delta} A^{\alpha\beta\gamma} = 0$ ($\forall \mu$) is equivalent
to the cyclic sum in the last 3 indices
vanishing.

$A^{\alpha\beta\gamma} + A^{\gamma\alpha\beta} + A^{\beta\gamma\alpha} = 0$
($\because$ contraction ~~of~~ with ϵ extracts 6 times totally anti sym part)
Hence,

$$A^{\alpha\beta\gamma} = \partial^{\alpha} F^{\beta\gamma}$$

$$\text{So, } \partial^{\alpha} F^{\beta\gamma} + \partial^{\beta} F^{\gamma\alpha} + \partial^{\gamma} F^{\alpha\beta} = 0$$

Again we relabel the indices ~~to~~ : $\alpha \rightarrow \lambda$

$\beta \rightarrow \mu$

$\gamma \rightarrow \nu$

$$\therefore \boxed{\partial^{\lambda} F^{\mu\nu} + \partial^{\mu} F^{\nu\lambda} + \partial^{\nu} F^{\lambda\mu} = 0}$$

$\hookrightarrow$ Bianchi
Identity

(3.) The relation $F^{\mu\nu}$ & its dual ~~$G^{\mu\nu}$~~ $G^{\mu\nu}$ is given
by: $G^{\mu\nu} = \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} F_{\alpha\beta} \rightarrow \textcircled{1}$

Now:

to express $F^{\mu\nu}$ in terms of $G_{\alpha\beta}$, we apply dual
operation & the double dual in Minkowski
space satisfies this identity:

$$(*)(*F)^{\mu\nu} = -F^{\mu\nu} \quad (\text{where } (*F)^{\mu\nu} = G^{\mu\nu})$$

dual operation

$$\Rightarrow \cancel{F}^{\mu\nu} = -(*G)^{\mu\nu} = -\frac{1}{2} \epsilon^{\mu\nu\lambda\sigma} G_{\lambda\sigma}$$

Now;

$$\underline{G}_{\lambda\sigma} = \frac{1}{2} \epsilon_{\lambda\sigma\lambda T} F^{\lambda T}$$

$$\frac{1}{2} \epsilon_{\mu\nu\lambda\sigma} G^{\lambda\sigma} = \frac{1}{2} \epsilon_{\mu\nu\lambda\sigma} \left(\frac{1}{2} \epsilon^{\lambda\sigma\lambda T} F_{\lambda T} \right)$$

The main contraction identity in Minkowski space is:

$$\epsilon_{\mu\nu\lambda\sigma} \epsilon^{\lambda\sigma\lambda T} = -2 \left(\delta_{\mu}^{\lambda} \delta_{\nu}^T - \delta_{\mu}^T \delta_{\nu}^{\lambda} \right)$$

(∵ $\epsilon_{\lambda\sigma\lambda T} = -\epsilon^{\lambda\sigma\lambda T}$)

so;

$$\frac{1}{4} \left[-2 \left(\delta_{\mu}^{\lambda} \delta_{\nu}^T - \delta_{\mu}^T \delta_{\nu}^{\lambda} \right) \right] F_{\lambda T} = -\frac{1}{2} (F_{\mu\nu} - F_{\nu\mu})$$

& ∴ $F_{\nu\mu} = -F_{\mu\nu}$

$$\Rightarrow -\frac{1}{2} (F_{\mu\nu} + F_{\mu\nu}) = -F_{\mu\nu}$$

Thus ;

$$\frac{1}{2} \epsilon_{\mu\nu\sigma} G^{\sigma} = -F_{\mu\nu}$$

(from eqn (ii))

Raising indices with the metric $\eta^{\mu\alpha} \eta^{\nu\beta}$ ($\because$ antisym preserves the form) gives :

$$F^{\mu\nu} = -\frac{1}{2} \epsilon^{\mu\nu\alpha\beta} G_{\alpha\beta}$$

(3) (a) $1 \text{ K} \rightarrow \text{GeV}$

(In natural units) $\Rightarrow \hbar = c = k_B = 1$ & Temp. (T) has dimensions of energy.

~~k_B~~ $\therefore E = k_B T$

$\rho \propto T^4$
 $\left[\frac{E}{V}\right] = [T]^4$
 $\left[\frac{\text{MeV}}{\text{fm}^3}\right] = [K]^4$
 $\left(\frac{0.197}{\text{MeV}}\right)^3$

$\Rightarrow T = \frac{E}{k_B}$

& $k_B = 8.6173 \times 10^{-5} \text{ eV/K}$ Kelvin

$1 \text{ K} = \frac{8.6173 \times 10^{-5} \text{ GeV}}{1 \text{ GeV} \cdot 10^9}$

$\Rightarrow [K] = [\text{MeV}]^{3/4} \times (0.197)^{3/4}$

$1 \text{ GeV} \cdot 10^9$

$[K] = (0.197)^{3/4} \text{ MeV}$

$1 \text{ K} = 8.6173 \times 10^{-14} \text{ GeV}$

(b) $1 \text{ Tesla} \rightarrow \text{GeV}^2$

In natural units, B has dimension of $(\text{energy})^2$.

$\mu = \frac{1}{2} \frac{B^2}{\mu_0}$

Energy density = $\frac{[E]}{[V]} = \frac{\text{GeV}}{(\text{fm})^3} = \frac{(\text{GeV})^4 \times (0.197)^3}{1} \left(\because 1 \text{ fm} = \frac{1}{0.197} \text{ GeV}^{-1} \right)$

$\Rightarrow (\text{GeV})^4 \times (0.197)^3 = \left[\frac{1}{2} \frac{[B]^2}{[\mu_0]} \right] = [B]^2$

$\Rightarrow T = [B] = \sqrt{\frac{2 \mu_0}{1}} (0.197)^{3/2} \text{ GeV}^2 \Rightarrow T = 0.0874 \text{ GeV}^2$